Review notes · for you

Vision and generation: unit check

University

Materials
Paper; The four lessons

Learning goals

- Four lessons
- One sitting

Activity: Work it

1. Answer all ten questions without notes.
2. Then read the explanation under each one, including the ones you got right.

Discussion

- Which lesson was the miss?

Check yourself

- Name the idea you will redo.

Try it this week
Redo the one item you missed, on paper, before the next course.

Quiz sheet

Vision and generation: unit check

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Name _____ Class _____ Date _____

1 **A blur filter sits on this patch. What is the new centre pixel?**

```
patch = [[9, 9, 9],
         [9, 0, 9],
         [9, 9, 9]]
# blur: all nine weights are 1/9
```

◯ 8 ◯ 9 ◯ 0

2 **In a trained vision network, people pick each filter's nine numbers by hand.**

◯ True ◯ False

3 **Pooling, as in the lab, keeps one number from this 2×2 block of filter responses. Which one?**

```
block = [[1, 5],
         [3, 2]]
```

◯ 5 ◯ 11 ◯ 2.75

4 **A detector tested only on faces shaped like its training faces scores 98%. That number says nothing about faces with other proportions.**

◯ True ◯ False

5 **A detector's line is 0.6. On a new test, close-set faces score 0.81, wide faces 0.52 and boxes 0.12. Which group does it miss?**

◯ The wide faces ◯ The boxes ◯ The close-set faces

6 **Two numbers that rebuild a blob well are a map of what the picture means.**

◯ True ◯ False

7 **An autoencoder squeezes a 64-pixel picture into a two-number code. What can the decode not bring back?**

◯ Details the two numbers never stored ◯ Nothing: the decode is exact ◯ Where the brightness sits

8 **Match the picture to how it was made.**

A picture rebuilt from a two-number code	• • • •	Diffusion
A sharp picture from a generator that learned to fool a discriminator	• • • •	Autoencoder
A picture that appears after many small denoise steps	• • • •	GAN

9 **With heavy noise, one denoise step can leave the picture further from the clean one than it started.**

◯ True ◯ False

10 **Order how a trained diffusion model makes a new picture.**

☐ Repeat the step many times until a picture appears

☐ Start from pure noise

☐ Take one learned denoise step

Check your answers

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1 **8**

The sum is 8 × 9 + 0 = 72, and 72 ÷ 9 = 8: the blur pulls the dark centre toward its bright neighbours.

2 **False**

Training learns the filter weights like any other weight.

3 **5**

This pooling keeps the strongest response, 5 (max pooling).

4 **True**

The test held only the shape it trained on, so a group it would miss was never asked about.

5 **The wide faces**

The wide faces are faces, and 0.52 is under the line, so they are missed.

6 **False**

Here they say where the brightness sits.

7 **Details the two numbers never stored**

The decode paints from the two numbers, so it can put the brightness in the right place.

8 **A picture rebuilt from a two-number code — Autoencoder**
A sharp picture from a generator that learned to fool a discriminator — GAN
A picture that appears after many small denoise steps — Diffusion

Rebuild, contest, denoise: three different losses, and none of the pictures is evidence.

9 **True**

In the lab, at full noise the difference grows from 0.28 to 0.33: one step pulls the noise to black or white and leaves speckles.

10 **Top to bottom: 3, 1, 2**

Training taught the step; making a picture repeats it, starting from noise.